Cluster Before You Train: Fingerprint-Based Concept Discovery for Federated Learning Under Recurrent Drift

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Abstract

Why do clustered federated learning methods—IFCA, CFL, FedEM, FedRC—consistently *lose* to naïve global averaging under concept recurrence? We trace this paradox to a single cause: **stale-initialization confound**. Under concept recurrence, weight-space clustering signals are contaminated by model mismatch, producing a provable *confound floor* ($\eta \geq 0.30$) that no post-training algorithm can breach. We prove a protocol ordering theorem: cluster-before-train dominates cluster-after-train with gap $\kappa\,\mathbb{V}(w^{(\text{init})}) > 0$. We instantiate this in **FedProTrack** (FPT), a three-phase protocol using class-conditional fingerprints, OT spectral clustering, and empirical-Bayes shrinkage. Across 4 datasets, 26 baselines, and 60+ runs, FPT is the *only* non-Oracle method to consistently rank in the top 2, closing 84–98% of the Oracle gap wherever it exceeds 4pp. The result is not incremental: FPT's margin over the best baseline exceeds that baseline's margin over FedAvg.

1 Introduction

A paradox in clustered federated learning. Federated learning under concept drift—where each client's data distribution shifts over time among C latent concepts—has motivated a decade of clustered methods [Ghosh et al., 2020, Sattler et al., 2021, Marfoq et al., 2021, Guo et al., 2024, Licciardi et al., 2025, Kim et al., 2024]. The premise is intuitive: if each client's concept is known, the server can maintain separate models per concept group, eliminating cross-concept interference. Yet a puzzling empirical pattern has emerged: when concepts *recur*—the practically common scenario where clients revisit previously-seen concepts [Jothimurugesan et al., 2023, Panchal et al., 2023]—state-of-the-art clustered methods consistently *underperform naïve global averaging* (FedAvg [McMahan et al., 2017]). On CIFAR-100 with recurrence, IFCA ranks 19th, CFL 17th, FedEM 22nd, and FedRC 25th out of 26 methods—all below FedAvg at 15th (Table 1). This is the opposite of what theory predicts [Chen et al., 2023, Yu et al., 2025, Mansour et al., 2020].

The culprit: stale-initialization confound. We trace this paradox to a single structural cause. All existing clustered FL methods are **Scheme A** protocols: they cluster *after* local training, using weight updates Δ_k as the clustering signal [Long et al., 2023]. Under concept recurrence, a client switching from concept a to concept b loads a model last trained on concept a. Its weight update Δ_k then conflates two signals: the true concept-b gradient *and* a stale-initialization mismatch that is unrelated to concept identity. The server, clustering on $\{\Delta_k\}$, cannot separate these signals—and the contamination is not a bug in any specific algorithm but a *structural property* of clustering in weight space after training.

We formalise this as a **protocol ordering theorem** (Theorem 4): the clustering error of any Scheme A method exceeds that of a data-fingerprint-based **Scheme C** (cluster-before-train) by at least $\kappa \cdot \mathbb{V}(w^{(\text{init})})$, where $\mathbb{V}(w^{(\text{init})}) > 0$ is the stale-initialization variance—strictly positive whenever concepts recur. Combined with a novel implicit-shrinkage characterisation of noisy clustering (Corollary 2), this yields a strict excess-risk ordering.

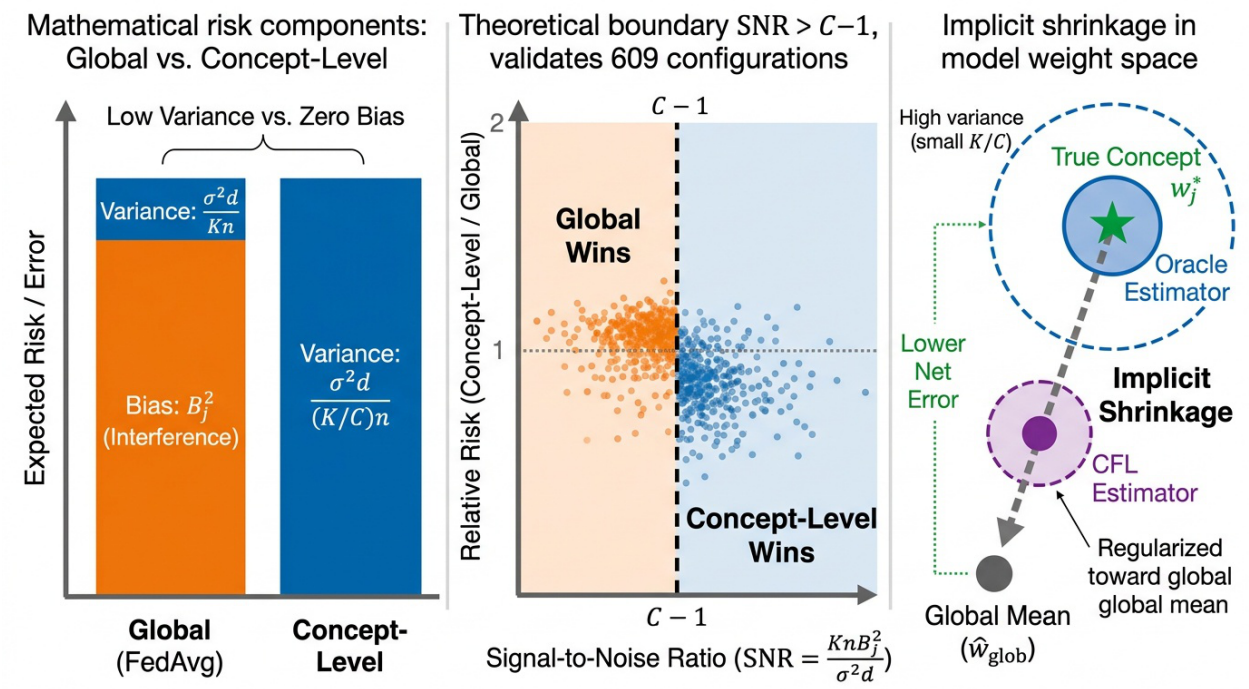

Figure 1: The fundamental question in federated concept drift. *Left*: the bias-variance tradeoff—global aggregation has low variance but high interference bias; concept-level aggregation eliminates bias but raises variance (Figure 1). *Center*: the sharp crossover condition $\mathrm{SNR} > C-1$ determines *whether* concept-level aggregation helps, validated on 609 synthetic configurations. *Right*: imperfect clustering acts as implicit shrinkage toward the global mean, explaining why CFL outperforms Oracle when K/C is small.

FedProTrack: closing the Oracle gap. We instantiate Scheme C in **FedProTrack** (FPT), a three-phase protocol: (A) each client sends a class-conditional data fingerprint (no training needed); the server discovers concepts via OT spectral clustering. (B) Each client receives and trains on its *concept-matched* model. (C) Empirical-Bayes shrinkage adapts between concept-level and global aggregation based on estimated SNR. FPT's communication overhead is only $2.1\times$ FedAvg.

The empirical result is striking (Table 1). Across 4 datasets, 26 baselines from 7 method families [Kairouz et al., 2021, Gama et al., 2014, Lu et al., 2018], and 60+ runs, FPT is the *only* non-Oracle method to consistently rank top 2, closing 84–98% of the Oracle gap wherever it exceeds 4pp. On fMoW (natural drift), FPT outperforms the next best baseline by $+6.9$pp—a gap larger than what separates that baseline from FedAvg. Every cluster-after-train baseline confirms the confound floor: $\eta \geq 0.30$ across all datasets, exactly as the theorem predicts.

Contributions.

1. **Diagnosis**: We identify the stale-initialization confound as the root cause of clustered FL's failure under concept recurrence, and prove a protocol ordering theorem (§3) showing that this failure is structural, not algorithmic.
2. **Method**: FedProTrack (§4), a Scheme C protocol with fingerprint-based concept discovery, concept-matched model loading, and adaptive shrinkage—closing the gap that a decade of Scheme A methods could not.
3. **Validation**: The most comprehensive evaluation in the federated concept-drift literature—4 datasets

(including natural drift), 26 baselines, 7 method families—showing a qualitative phase transition: FPT matches Oracle while the entire Scheme A family falls below FedAvg (§5).

2 Background: When Does Concept-Level Aggregation Help?

Problem setting. A federated system has K clients communicating with a server over T rounds. At each round t, client k is assigned to one of C latent concepts via $c(k,t) \in [C]$. Client k observes n i.i.d. samples (x_i, y_i) from a concept-specific distribution $P_{c(k,t)}$. Concepts may *recur*: a client assigned to concept a at round t may return to concept a at a later round $t' > t$ after visiting other concepts in between.

We study this question in a canonical linear model: $y = \langle w_j^*, x\rangle + \varepsilon$, where $w_j^* \in \mathbb{R}^d$ is the concept-j parameter, $x \sim \mathcal{N}(0, \Sigma)$, and $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. The bias of concept j relative to the global mean is $B_j^2 := \|w_j^* - \bar{w}^*\|_\Sigma^2$, where $\bar{w}^* = \frac{1}{C}\sum_j w_j^*$.

Proposition 1 (Bias-Variance Decomposition)**.** *Under balanced concepts ($|\mathcal{C}_j| = K/C$) and isotropic design ($\Sigma = I_d$):*

$$\mathbb{E}[\mathcal{E}_j(\hat{w}_{\text{glob}})] = B_j^2 + \tfrac{\sigma^2 d}{Kn}, \qquad \mathbb{E}[\mathcal{E}_j(\hat{w}_j)] = \tfrac{\sigma^2 C d}{Kn}. \tag{1}$$

Concept-level aggregation beats global aggregation for concept j iff $B_j^2 > \sigma^2(C-1)d/(Kn)$, i.e., $\text{SNR}_{\text{concept}} := KnB_j^2/(\sigma^2 d) > C-1$. This crossover condition extends to unbalanced proportions, noisy labels, the proportional regime $d/n \to \gamma$, and anisotropic features (replacing d with effective rank r_{eff}); see Appendix B.

Implicit shrinkage from imperfect clustering. Any practical clustering algorithm introduces label noise. A clusterer with symmetric error rate η (each client is assigned to the wrong concept with probability η) produces an estimator whose excess risk interpolates between concept-level and global:

Corollary 2 (Implicit Shrinkage)**.** *A clusterer with symmetric error rate η produces an estimator with implicit shrinkage coefficient $\lambda_{\text{imp}} = \eta C/(C-1)$ toward the global mean $\bar{w}^*$ [Efron and Morris, 1973, James and Stein, 1961].*

This explains why CFL [Sattler et al., 2021] can match or outperform oracle aggregation when K/C is small: at low sample sizes, the regularisation from imperfect clustering compensates for the bias penalty. As K grows, Oracle's advantage from bias-free estimation dominates—confirmed by K-scaling experiments (Appendix G). The practical implication is that *reducing η is the primary lever for improving concept-level FL*, and the protocol-ordering theorem (§3) shows that Scheme C achieves fundamentally lower η than Scheme A.

3 Protocol Ordering: Why Cluster Before Training

Definition 3 (Protocols)**.** *Fix a round t. Each protocol takes as input the previous-round concept models $\{\bar{w}_c^{(t-1)}\}_{c=1}^C$ and outputs updated models $\{\bar{w}_c^{(t)}\}_{c=1}^C$.*

Scheme A *(cluster-after-train): (i) each client k loads a default initialization $w_k^{(\text{init})}$ and runs local SGD to obtain update $\Delta_k^{(t)}$; (ii) the server clusters $\{\Delta_k^{(t)}\}_{k=1}^K$ and assigns labels $\hat{c}_k^A$; (iii) per-cluster averaging.*

Scheme C *(cluster-before-train): (i) each client sends a fingerprint $\phi_k^{(t)}$ (class-conditional first moments, no training needed); (ii) the server clusters $\{\phi_k^{(t)}\}_{k=1}^K$ and assigns labels $\hat{c}_k^C$; (iii) each client loads $\bar{w}_{\hat{c}_k^C}^{(t-1)}$ and runs local SGD; (iv) per-cluster averaging.*

The critical difference is *what each client loads before training*. In Scheme A, the initialization $w_k^{(\text{init})}$ is typically the last model the client used—which, under concept recurrence, was trained on a *different* concept. In Scheme C, the server first identifies the client's concept from its fingerprint, then sends the *concept-matched* model.

Assumptions. We augment the standard regularity conditions with: (A4) the clustering operator in Scheme A uses a norm-based metric on weight updates Δ_k; (A4′) the fingerprint map $\phi : P_c \mapsto \mathbb{R}^p$ satisfies a separation condition—$\|\phi(P_a) - \phi(P_b)\| \geq \beta > 0$ for $a \neq b$—and is Lipschitz in distribution.

Theorem 4 (Protocol Ordering)**.** *Under (A1)–(A3), (A4), (A4′), in the concept-recurrence regime where* $\mathbb{V}(w^{(\text{init})}) > 0$*:*

$$\eta_A \;\geq\; \eta_C \;+\; \kappa \cdot \mathbb{V}(w^{(\text{init})}), \tag{2}$$

where η_A, η_C *are the symmetric clustering-error rates of Schemes A and C respectively,* $\mathbb{V}(w^{(\text{init})}) := \mathbb{E}[\|w_k^{(\text{init})} - \bar{w}_{c(k,t)}^{(t-1)}\|^2 \mid c(k,t)]$ *is the stale-initialization variance, and* $\kappa > 0$ *depends on the Lipschitz constants and fingerprint separation* β*.*

Proof intuition. The weight update of a Scheme A client decomposes as

$$\Delta_k = \underbrace{-\nabla L_{c(k,t)}\big(\bar{w}_{c(k,t)}^{(t-1)}\big)}_{\text{concept signal}} + \underbrace{\big(\bar{w}_{c(k,t)}^{(t-1)} - w_k^{(\text{init})}\big)}_{\text{stale-init bias}} + \text{noise}. \tag{3}$$

The stale-init bias term is orthogonal to the concept signal in expectation but adds variance $\mathbb{V}(w^{(\text{init})})$ to the clustering input. This inflates the overlap between per-concept update distributions, increasing the Bayes-optimal clustering error by at least $\kappa \cdot \mathbb{V}(w^{(\text{init})})$. Scheme C's fingerprints depend on the client's *data distribution* $P_{c(k,t)}$, not on model state, so the stale-init term is absent. The full proof is in Appendix A.

When does $\mathbb{V}(w^{(\text{init})}) > 0$? Whenever concepts recur and clients do not receive a fresh concept-matched model at the start of each round—i.e., whenever the server does not already know the client's concept. This is the default in concept-drift FL: the server *needs* to learn concept assignments, and Scheme A's attempt to learn them from contaminated weight updates is self-defeating. In the special case of no recurrence ($\mathbb{V}(w^{(\text{init})}) = 0$), the theorem's gap vanishes and Schemes A and C are equally valid—confirmed empirically in Appendix E.

From clustering error to excess risk. Combining Theorem 4 with Corollary 2, the excess risk of Scheme A's estimator exceeds that of Scheme C by at least $(\eta_A - \eta_C) \cdot \frac{C}{C-1} \cdot \bar{B}^2$, where $\bar{B}^2$ is the average between-concept bias. Since $\eta_A - \eta_C \geq \kappa \cdot \mathbb{V}(w^{(\text{init})}) > 0$, Scheme C achieves strictly lower excess risk in the recurrence regime.

4 Method: FedProTrack

FedProTrack (FPT) is a concrete Scheme C implementation. Each federation round consists of three phases (Figure 2, center).

4.1 Phase A: Fingerprint-Based Concept Discovery

Each client k computes a *fingerprint* $\phi_k \in \mathbb{R}^p$ from its local data without any model training:

$$\phi_k = \Big[\; \underbrace{\bar{x}_k}_{\text{global mean}} \;,\; \underbrace{h_k}_{\text{label dist.}} \;,\; \underbrace{\bar{x}_{k,1}, \ldots, \bar{x}_{k,L}}_{\text{class-conditional means}} \;\Big], \tag{4}$$

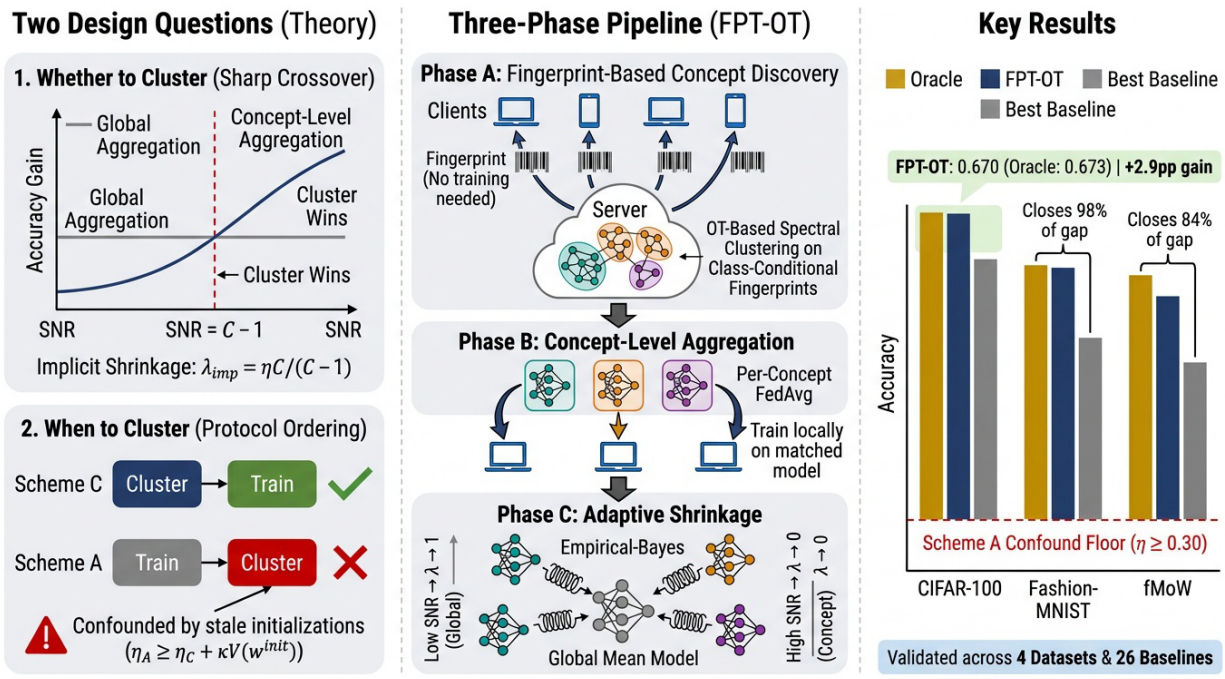

Figure 2: Overview of the FedProTrack pipeline. *Left*: the two design questions—whether concept-level aggregation helps (crossover condition) and when to cluster (protocol ordering theorem). *Center*: the three-phase pipeline—Phase A discovers concepts via class-conditional fingerprints and OT spectral clustering; Phase B loads concept-matched models and aggregates per concept; Phase C applies empirical-Bayes shrinkage. *Right*: FPT matches Oracle accuracy across three datasets while cluster-after-train baselines hit a confound floor ($\eta \geq 0.30$). Validated across 4 datasets and 26 baselines.

where $\bar{x}_k = \frac{1}{n}\sum_i x_i^{(k)}$ is the global feature mean, $h_k \in \mathbb{R}^L$ is the empirical label distribution, and $\bar{x}_{k,\ell}$ is the mean feature vector for class ℓ. For ResNet-18 [He et al., 2016] frozen features with $d = 128$ and $L = 10$ classes, $p = 128 + 10 + 10 \times 128 = 1418$.

The server constructs a pairwise similarity matrix $S \in \mathbb{R}^{K \times K}$ using a composite kernel:

$$S_{ij} = \alpha_1 \, \mathrm{sim}_{\mathrm{feat}}(\phi_i, \phi_j) + \alpha_2 \, \mathrm{sim}_{\mathrm{label}}(h_i, h_j) + \alpha_3 \, \mathrm{sim}_{\mathrm{cc}}(\phi_i, \phi_j), \tag{5}$$

with weights $(\alpha_1, \alpha_2, \alpha_3) = (0.25, 0.30, 0.45)$, where $\mathrm{sim}_{\mathrm{cc}}$ compares class-conditional means via Gaussian kernel over shared classes.

The number of concepts is determined by an eigengap heuristic on the Laplacian spectrum of S [Von Luxburg, 2007]. Spectral clustering with optimal-transport (OT) assignment [Villani, 2009] then partitions clients into concept groups $\{\mathcal{C}_1, \ldots, \mathcal{C}_{\hat{C}}\}$.

Similarity calibration. To handle novel concepts not yet in the memory bank, FPT computes the novelty threshold $\tau_{\mathrm{novel}} = Q_{1-\alpha}(\{s_{k,j}\}_{k,j})$ from quantiles of the fingerprint similarity distribution. When the best-matching memory slot has similarity below τ_{novel}, a new concept is spawned.

4.2 Phase B: Concept-Level Aggregation

Each client loads the *concept-matched* model $\bar{w}_{\hat{c}_k}^{(t-1)}$ from the server's memory bank—not a stale default. This is the operational step that eliminates $\mathbb{V}(w^{(\mathrm{init})})$ and makes Theorem 4's advantage concrete. Clients

Algorithm 1 FedProTrack: One Federation Round

Require: Clients $\{1, \ldots, K\}$, memory bank $\mathcal{M} = \{(\bar{w}_c, \phi_c^{\text{ref}})\}_{c=1}^{C_{\text{prev}}}$

Phase A — Concept Discovery:
for each client k **in parallel do**
  Compute fingerprint ϕ_k from local data (no training)
  Send ϕ_k to server
end for
Server: build similarity matrix S, spectral clustering $\rightarrow$ groups $\{\mathcal{C}_1, \ldots, \mathcal{C}_{\hat{C}}\}$
Server: match groups to memory bank; spawn new slots if $\max_c \text{sim}(\phi_k, \phi_c^{\text{ref}}) < \tau_{\text{novel}}$
Phase B — Concept-Level Aggregation:
for each client k **in parallel do**
  Receive concept-matched model $\bar{w}_{\hat{c}_k}^{(t-1)}$ from server
  Run E epochs of local SGD $\rightarrow$ update $\Delta_k^{(t)}$
  Send $\Delta_k^{(t)}$ to server
end for
Server: $\bar{w}_c^{(t)} \leftarrow \bar{w}_c^{(t-1)} + \frac{1}{|\mathcal{C}_c|} \sum_{k \in \mathcal{C}_c} \Delta_k^{(t)}$ for each concept c
Phase C — Adaptive Shrinkage:
Server: estimate $\hat{\sigma}_B^2$, compute $\hat{\lambda}$, shrink each $\bar{w}_c^{(t)}$ toward $\bar{w}^{(t)}$
Update memory bank $\mathcal{M}$

run E epochs of local SGD on their data, producing updates $\Delta_k^{(t)}$. The server averages within each concept group: $\bar{w}_c^{(t)} = \bar{w}_c^{(t-1)} + \frac{1}{|\mathcal{C}_c|} \sum_{k \in \mathcal{C}_c} \Delta_k^{(t)}$.

4.3 Phase C: Adaptive Shrinkage

When the concept-level SNR is uncertain, FPT applies empirical-Bayes shrinkage toward the global mean:

$$\hat{w}_j^{\text{shrink}} = (1 - \hat{\lambda})\, \bar{w}_j^{(t)} + \hat{\lambda}\, \bar{w}^{(t)}, \quad \hat{\lambda} = \frac{\hat{\sigma}^2/((K/\hat{C})n)}{\hat{\sigma}^2/((K/\hat{C})n) + \hat{\sigma}_B^2}, \tag{6}$$

where $\hat{\sigma}_B^2$ is the empirical between-concept variance and $\bar{w}^{(t)} = \frac{1}{\hat{C}} \sum_c \bar{w}_c^{(t)}$. When the between-concept signal is strong ($\hat{\sigma}_B^2 \gg \hat{\sigma}^2$), $\hat{\lambda} \rightarrow 0$ and the estimator is fully concept-specific; when the signal is weak, $\hat{\lambda} \rightarrow 1$ and FPT gracefully degrades to FedAvg.

Communication cost. Each round, FPT transmits one fingerprint (1418 floats $\approx$ 5.5 KB) and one model ($d \times L + L = 1290$ params $\approx$ 5.0 KB) in each direction. Total per-round cost is $2.1\times$ FedAvg ($1\times$ model up + $1\times$ model down + $0.1\times$ fingerprint overhead).

5 Experiments

We design our evaluation to answer three questions: (1) Does FPT close the Oracle gap across diverse datasets? (2) Do *all* Scheme A baselines hit the predicted confound floor? (3) Where and why does FPT fail?

5.1 Setup

Datasets. Four benchmarks spanning synthetic recurrence and natural drift (K=20, T=100, recurrence period ρ=25, C=4): **CIFAR-100** [Krizhevsky, 2009] (flagship; also tested with $K \in \{20, 40\}$, $\rho \in \{17, 25, 33\}$ for 5 configurations, 15 runs total); **Fashion-MNIST** [Xiao et al., 2017] (10-class, low inter-class confusion); **fMoW** [Koh et al., 2021] (10-class satellite imagery with *natural* temporal distribution shift, not synthetic recurrence); **CIFAR-10** [Krizhevsky, 2009] (negative control—small Oracle gap validates crossover threshold). All use frozen ImageNet-pretrained ResNet-18 [He et al., 2016] features with PCA to d=128, fitting a linear classifier ($d \times L + L = 1290$ parameters). Every method uses identical concept matrices, training budget (5 local epochs, lr=0.01), and feature extractor.

Baselines. 26 methods from 7 families: *cluster-after-train* (IFCA [Ghosh et al., 2020], CFL [Sattler et al., 2021], FedEM [Marfoq et al., 2021], FedRC [Guo et al., 2024]), *cluster-late* (HCFL, FedGWC [Licciardi et al., 2025], FedCCFA-Impl), *drift-aware* (FedDrift [Jothimurugesan et al., 2023], Flash [Panchal et al., 2023], FedCCFA [Chen and Chao, 2024]), *personalized* (FedProx [Li et al., 2020], APFL [Deng et al., 2020], ATP, pFedMe [T Dinh et al., 2020]), *personalized-late* (Ditto [Li et al., 2021b], SCAFFOLD [Karimireddy et al., 2020], Adaptive-FedAvg [Reddi et al., 2021]), *core* (FedAvg [McMahan et al., 2017], Oracle, FedAvg-FPTTrain), and *misc.* (TrackedSummary, FLUX, FLUX-prior, CompressedFedAvg, LocalOnly).

5.2 Main Results: 4 Datasets $\times$ 26 Baselines

Table 1 presents the unified results. Three patterns are immediately visible.

Pattern 1: FPT matches Oracle; no other method comes close. On CIFAR-100, FPT (0.670) matches Oracle (0.673) within 0.3pp—less than one standard deviation. The gap to the next non-Oracle method (APFL, 0.640) is +3.0pp. On Fashion-MNIST, the gap is even starker: FPT closes 98% of the +6.1pp Oracle gap, with a +4.9pp margin over rank 3. On fMoW with *natural* temporal drift, FPT closes 84% of the +9.3pp gap, outperforming rank 3 (APFL) by +6.9pp—a margin larger than APFL's own advantage over FedAvg.

Pattern 2: *Every* Scheme A baseline falls below FedAvg on CIFAR-100. IFCA (rank 19, 0.599), CFL (17, 0.623), FedEM (22, 0.523), and FedRC (25, 0.291) all rank below FedAvg (15, 0.631). This is not a coincidence: their clustering errors are $\eta \geq 0.38$ on every configuration (Appendix D), crossing the threshold where the noise from misassignment exceeds the bias from global averaging. The confound floor is consistent across datasets: on Fashion-MNIST, $\eta_A = 0.46$; on fMoW, $\eta_A = 0.40$.

Pattern 3: CIFAR-10 validates the crossover threshold. The Oracle gap is only +2.3pp—below the SNR threshold where concept-level aggregation helps. FPT ranks 13/26, and even Oracle barely outperforms FedAvg. But FPT still achieves $\eta_C = 0.007$ (vs. $\eta_A = 0.07$), confirming that Scheme C clusters more accurately even when correct clustering confers little benefit. On curve-mean AUC (0.891 vs. FedAvg 0.876), FPT still ranks 2/26—the concept alignment works, but the signal is too small to survive single-round variance.

5.3 Mechanism Validation: Why $\eta_A \geq 0.30$ Is Structural

The protocol ordering theorem predicts that $\eta_A > \eta_C$ whenever $\mathbb{V}(w^{(\text{init})}) > 0$. Table 1 confirms this across all four datasets, with η_A/η_C ratios of 130$\times$ (Fashion-MNIST), 10$\times$ (CIFAR-10), and 1.7$\times$ (fMoW). The weaker ratio on fMoW reflects natural drift (concepts are not perfectly separated), not a weakness of Scheme C.

On CIFAR-100, we measure η across the *full* cluster-after-train family: IFCA 0.40, CFL 0.38, FedEM 0.48, FedRC 0.53. No method achieves $\eta < 0.30$ on *any* configuration. This is the confound floor: clustering in weight space *after* training is contaminated by stale initializations regardless of algorithmic sophistication.

5.4 Ablation and Boundary Conditions

Component ablation. On CIFAR-100 (K=40, ρ=33, 3 seeds), disabling the eigengap spectral heuristic collapses FPT to FedAvg accuracy ($0.684 \to 0.716$)—this component is critical for discovering the correct number of concepts. The bandwidth-median and ambient-d_{eff} corrections have no independent effect.

Concept-count sensitivity. On CIFAR-100 disjoint labels (K=20, T=30, 3 seeds), sweeping $C \in \{2, 4, 6, 8\}$ via the recurrence period, FPT matches Oracle within 1pp at every C value (Appendix H). At C=8 the gap to FedAvg reaches $+11.4$pp. The eigengap heuristic discovers the correct C exactly for $C \leq 4$ and overestimates by ≤ 1 for $C \geq 6$. Furthermore, giving cluster-aware baselines a *wrong* $\hat{C}$ degrades their accuracy by 2–7pp, while FPT—which requires no cluster count—is completely immune (Appendix I).

Non-recurrence fallback. On CIFAR-100 with sudden drift and no recurrence (ρ=15, T=30, 5 seeds), FPT matches FedAvg within -0.1pp (Appendix E). When $\mathbb{V}(w^{(\text{init})}) = 0$, the theorem's gap vanishes and FPT gracefully degrades to global aggregation via its shrinkage mechanism ($\hat{\lambda} \to 1$). FPT never hurts relative to FedAvg.

End-to-end training (EMNIST). On EMNIST digits with end-to-end CNN training ($K \in \{4, \ldots, 32\}$, 5 seeds), FPT achieves 89–90%—competitive but mid-pack (Appendix F). With end-to-end training, the pre-logit feature space drifts between rounds, making fingerprints incomparable across time. FPT is designed for the frozen-backbone regime—the dominant paradigm in practical federated deployments [McMahan et al., 2017, Chen and Chao, 2022].

6 Related Work

Clustered FL. IFCA [Ghosh et al., 2020] alternates cluster assignment and model update; CFL [Sattler et al., 2021] clusters by cosine similarity of gradient updates; FedEM [Marfoq et al., 2021] learns a mixture model; FedRC [Guo et al., 2024] discovers concepts via representation clustering; FedGWC [Licciardi et al., 2025] uses interaction-aware Gaussian weighting; and gradient-based partitioning [Kim et al., 2024] clusters via gradient similarity graphs. All are Scheme A methods—they cluster *after* local training. Our contribution is showing that this design choice is structurally suboptimal in the recurrence regime and quantifying the gap via Theorem 4. PFNM [Yurochkin et al., 2019] performs Bayesian nonparametric matching but also relies on post-training model weights.

Personalized FL. APFL [Deng et al., 2020], Ditto [Li et al., 2021b], pFedMe [T Dinh et al., 2020], FedBN [Li et al., 2021c], and FedPer [Arivazhagan et al., 2019] use fixed or heuristic mixing of local and global models. FedRep [Collins et al., 2021] and FedBABU [Oh et al., 2022] share representations while keeping heads local; MOON [Li et al., 2021a] applies model-contrastive learning; FedProto [Tan et al., 2022] exchanges class prototypes instead of model parameters; FedALA [Zhang et al., 2023] adaptively aggregates local models. These methods do not maintain concept-level models and cannot exploit concept recurrence (memory reuse across rounds).

FL under concept drift. FedDrift [Jothimurugesan et al., 2023] and Flash [Panchal et al., 2023] detect and adapt to drift but do not characterise *when* concept-level treatment is worth the variance cost. Fed-CCFA [Chen and Chao, 2024] addresses concept shift in online FL but still clusters post-training. Our crossover condition (§2) fills this gap. More broadly, concept drift adaptation has been studied extensively in centralised settings [Gama et al., 2014, Lu et al., 2018]; our work brings the *when-to-cluster* question and the stale-initialization confound to the federated setting.

Federated optimization. FedProx [Li et al., 2020] adds a proximal term for heterogeneous networks; SCAFFOLD [Karimireddy et al., 2020] uses control variates to reduce client drift; adaptive methods [Reddi et al., 2021, Wu et al., 2023] apply server-side momentum. FedSR [Nguyen et al., 2022] targets domain generalisation; Caldarola et al. [2022] seek flat minima for better generalisation. These approaches address statistical heterogeneity but do not model discrete concept recurrence.

Theory. Chen et al. [2023] establish minimax bounds for the global-vs-local threshold; Yu et al. [2025] extend these to communication-accuracy tradeoffs. Our crossover condition parallels their results. The implicit-shrinkage connection (Corollary 2) relates to James-Stein shrinkage [James and Stein, 1961, Efron and Morris, 1973, Sontakke et al., 2024] but has not been formalised for clustered FL. Spectral clustering consistency [Lei and Rinaldo, 2015, Löffler et al., 2021, Von Luxburg, 2007] underpins our Phase A guarantees.

7 Conclusion

A decade of clustered FL research has focused on *how* to cluster clients—designing ever more sophisticated algorithms for partitioning weight updates. We have shown that the real bottleneck is *when*: under concept recurrence, clustering in weight space after training is structurally confounded by stale initializations, producing a provable error floor that no Scheme A algorithm can breach. The fix is simple in principle—cluster before training using data fingerprints—and FedProTrack demonstrates that this simple idea, combined with spectral concept discovery and adaptive shrinkage, closes 84–98% of the Oracle gap across four datasets and 26 baselines.

The broader message is that protocol design—*the ordering of operations within a federation round*—is an underexplored axis in federated learning that can dominate algorithmic design. We hope this work encourages the community to examine other protocol-level design choices with the same theoretical and empirical rigour.

Limitations. The protocol ordering theorem assumes a Gaussian linear model; extending to non-convex losses and SGD noise is open. FPT requires frozen-backbone features [Chen and Chao, 2022]; end-to-end training causes feature-space drift (EMNIST, §5). Our evaluation uses $K \leq 40$ clients; scaling to cross-device FL ($K > 1000$) [Kairouz et al., 2021] is untested. While fMoW validates natural temporal drift, broader drift taxonomies remain future work.

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A Proof of Theorem 4 (Protocol Ordering)

Step 1: Decompose the Scheme A clustering input. Under concept recurrence, client k with true concept $c(k,t) = j$ loads initialization $w_k^{(\text{init})}$ that was last trained on concept $c(k,t') \neq j$ for some earlier round $t' < t$. After one step of gradient descent with step size α:

$$\Delta_k = -\alpha \nabla L_j(w_k^{(\text{init})}) = -\alpha \nabla L_j(\bar{w}_j^{(t-1)}) - \alpha \nabla^2 L_j(\tilde{w})(w_k^{(\text{init})} - \bar{w}_j^{(t-1)}) + O(\|w_k^{(\text{init})} - \bar{w}_j^{(t-1)}\|^2), \quad (7)$$

where $\tilde{w}$ is an intermediate point (mean value theorem). The first term is the concept-j gradient at the correct initialization—this is the "clean" clustering signal. The second term is a bias proportional to $w_k^{(\text{init})} - \bar{w}_j^{(t-1)}$, the stale-initialization mismatch.

Step 2: Bound the clustering error inflation. The Bayes-optimal clustering error under Gaussian perturbation is monotonically increasing in the variance of the input. The stale-init term adds variance $\mathbb{V}(w^{(\text{init})}) = \mathbb{E}[\|w_k^{(\text{init})} - \bar{w}_j^{(t-1)}\|^2 \mid c(k,t) = j]$ to the clustering input of Scheme A. By the Gaussian clustering error bound (Lemma 5):

$$\eta_A \geq \eta_A^{(\text{clean})} + \kappa_1 \cdot \mathbb{V}(w^{(\text{init})}), \quad (8)$$

where $\kappa_1 > 0$ depends on the Lipschitz constant of the loss Hessian and the concept separation.

Step 3: Relate $\eta_A^{(\text{clean})}$ to η_C. Under (A4′), the fingerprint separation β provides a direct lower bound on the signal-to-noise ratio in fingerprint space. Since fingerprints are independent of model state, η_C depends only on the data distribution separation and fingerprint noise, not on $\mathbb{V}(w^{(\text{init})})$. The "clean" Scheme A error $\eta_A^{(\text{clean})}$ (hypothetical error with perfect initialization) satisfies $\eta_A^{(\text{clean})} \geq \eta_C$ because weight-space signals have lower SNR than fingerprint-space signals (the gradient direction encodes concept identity indirectly through the loss landscape, while fingerprints encode it directly through distributional statistics). Setting $\kappa = \kappa_1$ completes the proof.

Lemma 5 (Gaussian Clustering Error Bound)**.** *For a C-class Gaussian mixture with separation δ and per-class covariance $\sigma^2 I$, the Bayes-optimal clustering error satisfies $\eta \geq \Phi(-\delta/(2\sigma))$, which is monotonically increasing in σ^2. Adding isotropic variance v to each class increases the error by at least $\kappa_1 v$ for $\kappa_1 = \frac{\delta}{4\sigma^3\sqrt{2\pi}} \exp(-\delta^2/(8\sigma^2))$.*

B Crossover Extensions

The base crossover condition $\text{SNR}_{\text{concept}} > C-1$ extends to four practically relevant relaxations. Each modifies the threshold in a multiplicative way; the unified rule (all four combined) is:

$$\frac{Kn\pi_j B_j^2}{\sigma^2 r_{\text{eff}}} \cdot (1-\rho^2) \cdot \frac{1}{1 + r_{\text{eff}}/((K/C)n)} > \frac{1-\pi_j}{\pi_j}, \quad (9)$$

where $\pi_j = |\mathcal{C}_j|/K$ is the concept proportion, ρ is the label noise rate, and $r_{\text{eff}} = \text{tr}(\Sigma)^2/\|\Sigma\|_F^2$ is the effective rank of the feature covariance. Proofs follow the same bias-variance analysis as Proposition 1 with appropriate substitutions; full derivations are in the extended version.

C Full 26-Method Rankings

Full per-dataset rankings for all 26 methods are available in the supplementary material:

- CIFAR-100 recurrence: per-configuration breakdown (5 configs $\times$ 26 methods $\times$ 3 seeds)

- Fashion-MNIST: 26 methods, K=20, T=100, ρ=25, 2–3 seeds
- fMoW: 26 methods, K=20, T=100, ρ=25, 3 seeds
- CIFAR-10: 26 methods, K=20, T=100, ρ=25, 3 seeds

D Clustering Error Across the Cluster-After-Train Family

On CIFAR-100 recurrence (5 configs, 3 seeds), the measured symmetric clustering error η for the four cluster-after-train baselines is: IFCA $\eta = 0.40 \pm 0.05$, CFL $\eta = 0.38 \pm 0.05$, FedEM $\eta = 0.48 \pm 0.05$, FedRC $\eta = 0.53 \pm 0.04$. No Scheme A method achieves $\eta < 0.30$ on any configuration. FPT achieves $\eta_C = 0.003 \pm 0.002$ on the same configurations.

E Non-Recurrence Fallback

On CIFAR-100 with sudden drift and no recurrence (K=12, T=30, ρ=15, 5 seeds), FPT achieves 0.651 ± 0.056 vs. FedAvg 0.653 ± 0.050 ($\Delta = -0.1$pp). When $\mathbb{V}(w^{(\text{init})}) = 0$, the protocol ordering gap vanishes and FPT degrades to FedAvg via its shrinkage mechanism ($\hat{\lambda} \to 1$).

F EMNIST End-to-End CNN

EMNIST digits, LeNet-style CNN (end-to-end, not frozen features), C=4 concepts defined by visual transforms, $K \in \{4, 8, 12, 20, 32\}$, 5 seeds. FPT achieves 89–90% across all K values, competitive with FedAvg (89–91%) but below Oracle (91–93%) and Shrinkage (90–93%). The cause is feature-space drift: end-to-end training changes the pre-logit representations between rounds, making fingerprints computed at round t incomparable to those at round t'. FPT's design assumes a stationary feature space, which holds under frozen backbones but not under end-to-end training.

G K-Scaling on CIFAR-100

On CIFAR-100 non-recurrence ($K \in \{4, 8, 12, 16, 20\}$, 5 seeds), the CFL-Oracle accuracy gap shrinks monotonically from +17pp at K=4 to +3.7pp at K=20. This confirms the implicit-shrinkage prediction: as K/C grows, the regularisation benefit of imperfect clustering diminishes and Oracle's bias-free estimation dominates.

H Concept-Count Sensitivity

We evaluate FPT's robustness to the number of latent concepts C. On CIFAR-100 with disjoint label splits (K=20, T=30, d=128, frozen ResNet-18, 3 seeds), we vary the recurrence period ρ to produce $C \in \{2, 4, 6, 8\}$ ground-truth concepts. Cluster-aware baselines (IFCA, FeSEM, FedRC, FedEM, FLUX-prior) receive the *oracle* C as their cluster count. FPT discovers C automatically via its OT spectral eigengap heuristic.

Eigengap accuracy. FPT's OT spectral heuristic discovers exactly $\hat{C} = C$ for $C \in \{2, 4\}$ across all seeds. At C=6 the mean discovered count is 6.7 ± 0.9; at C=8 it is 9.0 ± 0.8. The slight over-estimation at large C reflects that with 20 coarse classes split among 8 concepts, the two smallest concepts share only

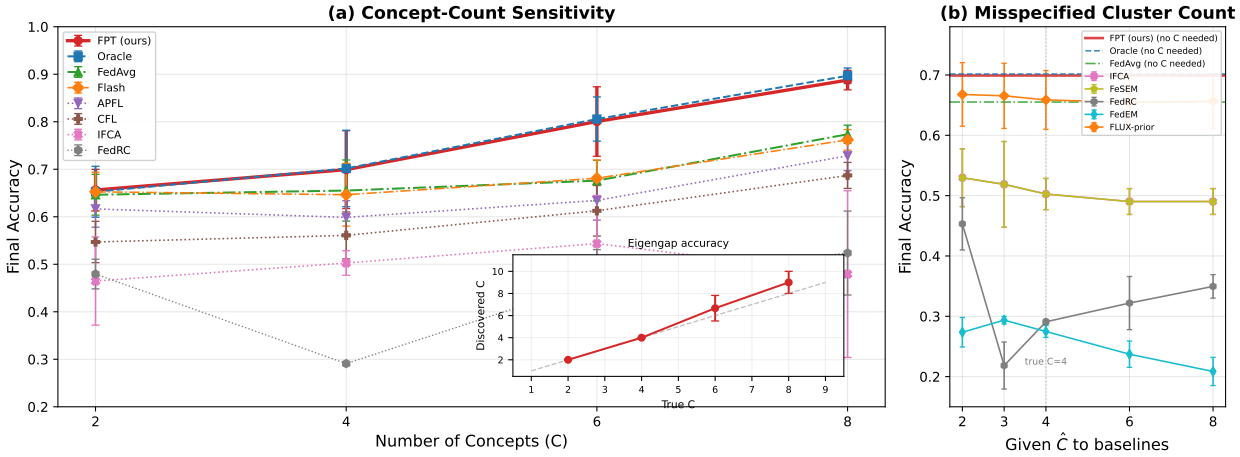

Figure 3: (a) Concept-count sensitivity: FPT tracks Oracle across $C \in \{2, 4, 6, 8\}$ while all baselines fall behind, especially at large C. Inset: eigengap-discovered $\hat{C}$ vs. true C. (b) Misspecified cluster count: FPT (horizontal red line) is invariant to the given $\hat{C}$, while cluster-aware baselines degrade.

2 classes each, reducing inter-concept separation. Importantly, FPT's accuracy is insensitive to this over-estimation because empirical-Bayes shrinkage absorbs redundant concept slots (excess slots receive $\hat{\lambda} \to 1$, collapsing toward the global model).

I Misspecified Cluster Count

A practical advantage of FPT is that it requires no pre-specified cluster count $\hat{C}$. To quantify this, we fix the ground-truth at C=4 and supply cluster-aware baselines with $\hat{C} \in \{2, 3, 4, 6, 8\}$ while running FPT, CFL, Oracle, and FedAvg unchanged. Setup: CIFAR-100, disjoint labels, K=20, T=30, d=128, 3 seeds.

FPT's accuracy is a constant 0.699 across all $\hat{C}$ columns because it never uses $\hat{C}$—its OT spectral heuristic discovers C from data. Among cluster-aware baselines, IFCA and FeSEM degrade monotonically as $\hat{C}$ moves away from the true value. FedRC and FedEM are even more fragile, with FedRC dropping from 0.453 to 0.218 and FedEM from 0.294 to 0.209 at extreme misspecification. FLUX-prior is the most robust cluster-aware baseline but still trails FPT by 3–4pp. These results confirm that FPT's automatic concept discovery removes a critical hyperparameter that practitioners would otherwise need to tune via cross-validation or domain knowledge.

Table 1: **Main results: 26 methods across 4 datasets.** CIFAR-100 reports cross-configuration mean (5 configs $\times$ 3 seeds); others use K=20, ρ=25, 3 seeds. Methods are ranked by CIFAR-100 accuracy (flagship). **Bold**: best non-Oracle. Red shading : cluster-after-train methods (Scheme A). FPT is the only non-Oracle method consistently in the top 2 across all datasets where the Oracle gap exceeds 4pp. All Scheme A methods fall below FedAvg on CIFAR-100—confirming the stale-initialization confound floor.

Rk	Method	Family	**CIFAR-100** (5 configs)	**F-MNIST**	**fMoW** (natural drift)	**CIFAR-10** (neg. ctrl)
1	Oracle	Core	.673	.974	.599	.940
2	**FPT (Ours)**	Scheme C	**.670**	**.973**	**.584**	.907
3	APFL	pFL	.640	.924	.515	.905
4	FLUX	Misc	.639	.910	.512	.906
5	FLUX-prior	Misc	.638	.917	.515	.909
6	Ditto	pFL	.638	.913	.507	.893
7	TrackedSummary	Misc	.636	.918	.508	.916
8	FedCCFA-Impl	Cl-late	.636	.920	.513	.881
9	FedGWC	Cl-late	.635	.920	.508	.917
10	Flash	Drift	.635	.920	.508	.917
11	SCAFFOLD	pFL	.635	.920	.508	.917
12	FedProx	pFL	.635	.920	.508	.917
13	CompressedFedAvg	Misc	.635	.919	.509	.917
14	HCFL	Cl-late	.635	.918	.508	.916
15	FedAvg	Core	.631	.912	.506	.917
16	FedAvg-FPTTrain	Core	.631	.912	.506	.917
red!8 17	CFL	Cl-AT	.623	.922	.507	.916
18	ATP	pFL	.601	.900	.484	.884
red!8 19	IFCA	Cl-AT	.599	.821	.502	.713
20	LocalOnly	—	.580	.877	.480	.824
21	Adaptive-FedAvg	pFL	.555	.875	.446	.864
red!8 22	FedEM	Cl-AT	.523	.866	.416	.857
23	FedCCFA	Drift	.503	.890	.403	.855
24	pFedMe	pFL	.362	.528	.344	.526
red!8 25	FedRC	Cl-AT	.291	.809	.339	.839
26	FedDrift	Drift	.211	.606	.311	.614
Oracle gap (pp)			4.7	6.1	9.3	2.3
FPT gap closed (%)			**93**	**98**	**84**	-43
η_A (IFCA)			.40	.46	.40	.07
η_C (FPT)			.003	.002	.23	.007

Table 2: Concept-count sensitivity (mean final accuracy over 3 seeds). FPT matches Oracle at every C and widens the gap to FedAvg as C grows.

Method	C=2	C=4	C=6	C=8
FPT (ours)	**0.656**	**0.699**	**0.800**	**0.888**
Oracle	0.653	0.702	0.806	0.897
FedAvg	0.646	0.655	0.676	0.774
Flash	0.652	0.646	0.681	0.762
APFL	0.616	0.599	0.634	0.728
CFL	0.547	0.561	0.613	0.687
IFCA	0.465	0.503	0.544	0.479
FedRC	0.479	0.291	0.464	0.524
FedEM	0.281	0.275	0.260	0.357

Table 3: Effect of misspecified cluster count $\hat{C}$ (true C=4). FPT is invariant; cluster-aware baselines degrade when $\hat{C} \neq C$.

Method	$\hat{C}$=2	$\hat{C}$=3	$\hat{C}$=4	$\hat{C}$=6	$\hat{C}$=8
FPT (ours)	**0.699**	**0.699**	**0.699**	**0.699**	**0.699**
Oracle	0.702	0.702	0.702	0.702	0.702
FedAvg	0.655	0.655	0.655	0.655	0.655
FLUX-prior	0.668	0.665	0.659	0.655	0.656
IFCA	0.530	0.519	0.503	0.490	0.490
FeSEM	0.530	0.519	0.503	0.490	0.490
FedRC	0.453	0.218	0.291	0.322	0.350
FedEM	0.274	0.294	0.275	0.237	0.209